

Field Test Procedure

16MAR26

Document Information

Document Title	Field Test Procedure
Document ID	TP-001
Version	1.7
Author(s)	OP, GR, MS, HK
Approver(s)	*Whomever*
Date	16MAR26

Revision

Version	Date	Author	Description of Changes
1.0	27FEB26	OP	Initial Draft
1.1	28FEB26	OP	Adjustments to Format/Test Procedure
1.2	01MAR26	GR	Modified the procedure to include mass measurements
1.3	03MAR26	MS, HK	Adding damage modes and sweeps
1.4	03MAR26	HK	Changing to Field Test: altering procedure and maneuvers to reflect field test
1.5	13MAR26	MS	Changed "and or" to and/or, page 6 test procedure 6.
1.6	13MAR26	OP	Changed title
1.7	16MAR26	HK	Updated Dive Set and Fault Cases

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1 Motivation

Field testing will confirm the digital twin's data outputs, as well as determine the percent effect of variable wing and rudder damage on those data outputs. This will both finalize critical system and functional requirement requirement data needs, as well as begin the characterization of wing and rudder damage in the fault database.

2 Scope

The intention is for this test to occur in the first week of UW Spring Quarter 2026, date and time to be finalized at the UW Student Seaglider Center's (SSC) discretion. It is intended to integrate with SSC operations and minimize the impact on SSC scheduling. As such, the field test will resemble their normal operational procedures with the only addition being the enacted wing and rudder fault cases.

3 Test Environment/Safety/

i Test Environment

The field test will occur through deploying the Seaglider off the boat's launch cradle and into the open Puget Sound, where the Seaglider will dive as it would during a normal operation. Up until deployment, all procedures, precautions, and scheduling will be under the jurisdiction of the SSC.

ii Safety

Due to the nature of open ocean boat travel and Seaglider mission deployment, any capstone member selected to overview the field test must acquire the proper training under the SSC's discretion. All UW School of Oceanography and SSC instructions will be followed in order to prevent risk of injury or death.

4 Roles and Responsibilities

While the following roles are required to operate the test efficiently, personnel constraints will limit the amount of these roles that will be on the boat for deployment. The rest of the testing personnel shall attend to the pilots and observe the test remotely from the mission control center.

1. _____ **Field Overseer:** Oversees and assists with the field test to ensure proper following of testing procedures, as well as provide a first hand account of the field test. This is the only personnel 100% required to be on the boat for the test to be carried out.
2. _____ **Videographer:** Collects photographic and video data for later analysis using camera of choice. This role can be carried out by the field overseer for the boat operation, as well as an individual assigned to mission control for the piloting portion.
3. _____ **Test Lead:** Coordinates with SSC and organizes Capstone members. This role has no locational requirement.
4. _____ **Data Lead:** Collects Seaglider data from SSC team for analysis. This role will be filled by an individual at mission control.
5. _____ **Test Members:** Any Capstone members assisting with or observing test

5 Test Procedure

i Equipment

- ☐ Laptop
- ☐ Camera
- ☐ Data Recording Device (laptop/USB Drive)
- ☐ Note Taking Device
- ☐ Scale
- ☐ Closed Toe Shoes (for crush hazards)
- ☐ Hardhats *as required*
- ☐ Life vests *as required*
- ☐ SSC chosen wing/rudder cutting equipment *as required*
- ☐ Appropriate breathing equipment for chosen cutting method *as required*

ii Seaglider Control Limits

Sweep	Max	Min	Center
Pitch	3.3 cm (aft)	-8.3 cm (fwd)	2850 (AD)
Roll	50° (stbd)	-50° (port)	1991 (AD)
VBD	548.2 (cc)	-300.5 (cc)	2735

Table 1: Min, Max, and Center position for SG 195

iii Test Scenarios and Damage Modes

1 Nominal (No Damage Modes/Fault Cases)

1. 100% of wing and rudder attached

2 Off Nominal (Damage Modes/Fault Cases)

1. 50.0% of port side wing span remaining
2. 0% of top rudder span remaining

3 Test Dives

All nominal and off nominal cases will perform this test dive procedure. The dive is to reflect a typical Seaglider operational dive set.

1. Dive 1: 120 meter dive depth
2. Dive 2: 120 meter dive depth

All other dive input parameters are determined by the pilot in order to represent a typical Seaglider mission given these depths. For each dive, the mission parameters and inputs are to be recorded, tabulated, and kept consistent for each fault case undergoing this dive set. These dive depths can be changed upon SSC piloting discretion or recommendations.

iv Procedure

1. _____ Ensure pre-test mass measurements are recorded, as requested by capstone Kinematics and Dynamics team:

Mass of Seaglider (kg)	
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2. _____ Using the scale weigh the Seaglider control surfaces and record their masses.

Control Surface	Mass (kg)
Wing 1	
Wing 2	
Rudders	

3. _____ Install rudders and wings according to the standard configuration of the Seaglider
4. _____ Using the scale, weigh the Seaglider with all control surfaces attached and record the total mass

Total Mass of Seaglider (kg)	
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5. _____ Repeat steps 1-4 for each damage case if damage case wing and rudders are prepared prior to the test day. If fault specimens are created actively on the boat, continue to step 6. Confer with SSC if fault specimens can be weighed on the boat.
6. _____ Confirm with SSC staff that measurements and/or measurement plans are complete.
7. _____ Follow SSC field-day schedule up until final glider assembly on the boat, where the testing procedure will begin.
8. _____ Prepare Seaglider for full assembly once deployment point is reached and assembly is commanded.
9. _____ Attach nominal wing and rudder specimens to the Seaglider, along with all other nominal components.
10. _____ Ready nominal Seaglider on deployment cradle and await deployment instruction.
11. _____ Deploy nominal Seaglider and transfer control to pilots upon completion of SSC Field Team checks.

12. _____ Seaglider performs mission procedure as given by pilots and informed by the requested dive profile set.
13. _____ Perform SSC retrieval procedure and disassemble nominal wing and rudder glider specimens.
14. _____ Repeat steps 9-13 for all damage modes, attaching pre-made fight surface specimens, or creating fault specimens on site under SSC discretion.
15. _____ Follow SSC return procedure once final fault case, dive, and retrieval has concluded.
16. _____ Ensure all dive data has been collected and appropriate for the transfer from the SSC science team to the capstone team.
17. _____ Assist SSC or Pilot Staff with clean-up as requested for field and piloting operations
18. _____ Ensure all Capstone equipment is stowed and secured
19. _____ Exit field and mission control premises and upload data

v Expected Results

It is expected that the Seaglider will perform rolls, pitches, and buoyancy maneuvers differently with damaged surfaces than without. The damage cases are expected to cause higher rates of oscillatory roll and pitch maneuvers in the given test sections. Other expected observations in damage cases include abnormal data changes from Seaglider orientation disturbances in otherwise unaffected data sets. Examples of this include only buoyancy data changing in a nominal buoyancy maneuver, but both buoyancy and roll data change in an off-nominal buoyancy maneuver. What data is affected and the exact percentage difference is to be determined for each fault individually from it's collected data file.

vi Actual Results

Affected Item	% Control Surface Span Remaining	Mass (kg)	Expected Results	Notes	P/F
Port Side Wing / Top Rudder	100% Remaining - Nominal		Data and video of acceptable quality and test cleared for simulating damage.		☐ P ☐ F
Port Side Wing	50% Remaining - Off Nominal		Maneuvers cause detectable orientation shift.		☐ P ☐ F
Top Rudder	0% Remaining - Off Nominal		Maneuvers cause detectable orientation shift.		☐ P ☐ F

6 Acceptance Criteria

1. (Yes/No) Data collected showed noticeable difference in nominal/off-nominal conditions
2. (Yes/No) Video post-analysis confirms data collected
3. (Yes/No) Test was concluded safely

4. (Yes/No) SSC pilots are capable of reproducing maneuvers from test in real-world scenarios

7 Test Acceptance

By signing below, the UW AA Ocean Capstone Test Lead and Approver confirm that the test procedure was executed in the designated environment

Test Lead Signature & Date

Approver Signature & Date

8 Appendix

i Test Data

ii Glossary

iii Reference